

RWANDA TVET BOARD
MARKING GUIDE — CONFIDENTIAL
[FORMATIVE ASSESSMENT]

Module: Backend Application Development Using Node JS | **Program:** SOFTWARE DEVELOPMENT |
Level: 4 | **Total Marks:** 45

Q 1.	Analyze the importance of Node.js in developing RESTFUL APIs with Node JS. Identify advantages and limitations.	(7 marks)
Answer: Key steps: 1. Node.js: 2. NPM (Node Package Manager): 3. Dependencies: Commands: APIs let you incorporate additional functionality into your application, which can delivery application, you might incorporate a third-party mapping API to let users Award marks for correct explanation of Node.js in context. Marking guide: Award 7 marks for correct and complete answer.		
Q 2.	Match each term in Column A with its correct description in Column B.	(4 marks)
Answer: 1. Node.js 2. A backend application 3. methods. It 4. An object in JavaScript Marking guide: Award 4 marks for correct and complete answer.		
Q 3.	What is the main purpose of MongoDB when developing RESTFUL APIs with Node JS?	(2 marks)
Answer: B Marking guide: Award 2 marks for correct and complete answer.		
Q 4.	Mongoose directly affects the security of developing RESTFUL APIs with Node JS.	(2 marks)
Answer: True Marking guide: Award 2 marks for correct and complete answer.		
Q 5.	Which design using JWT would best support developing RESTFUL APIs with Node JS?	(2 marks)
Answer: D Marking guide: Award 2 marks for correct and complete answer.		
Q 6.	Explain how bcrypt works in the context of developing RESTFUL APIs with Node JS.	(8 marks)
Answer: Key steps: 1. You are requested to discuss the following: 2. Provide your findings and write them on paper. 3. Present the findings to the whole class Commands: 7. Promises ES6: Introduces Promises, providing a cleaner way to handle asynchronous Award marks for correct explanation of bcrypt in context. Marking guide: Award 8 marks for correct and complete answer.		
Q 7.	Demonstrate how to use REST API when developing RESTFUL APIs with Node JS. Provide a step-by-step explanation.	(8 marks)

Answer: Key steps: 1. Read key reading 1.3.1 | 2. Referring to the previous practical activities (1.2.4.) you are requested to go to the | 3. Apply safety precautions.
Commands: `npm install mysql2` | `const mysql = require('mysql2');` Award marks for correct explanation of REST API in context.

Marking guide: Award 8 marks for correct and complete answer.

Q 8. Understanding RESTful API is required before developing RESTFUL APIs with Node JS. **(2 marks)**

Answer: True

Marking guide: Award 2 marks for correct and complete answer.

Q 9. Evaluate the effectiveness of Middleware when developing RESTFUL APIs with Node JS. Justify with examples. **(8 marks)**

Answer: Key steps: 1. Node.js: | 2. NPM (Node Package Manager): | 3. Dependencies: Commands: APIs let you incorporate additional functionality into your application, which can | delivery application, you might incorporate a third-party mapping API to let users Award marks for correct explanation of Middleware in context.

Marking guide: Award 8 marks for correct and complete answer.

Q 10. How would you correctly apply Authentication when developing RESTFUL APIs with Node JS? **(2 marks)**

Answer: A

Marking guide: Award 2 marks for correct and complete answer.